University of Creative Technology, Chittagong (UCTC)
Department of Computer Science and Engineering
CSE 6th Semester, R1, 13th Batch — Artificial Intelligence Lab

AI-Driven Smart Rain and Water-Level Monitoring & Alert System

Using ESP32 and ThingSpeak

Submitted to: *Prof. Dr. Mohammad Shahadat Hossain*
Course Instructor and Academic Advisor

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1. Introduction

We are thrilled to unveil our Artificial Intelligence Lab project for the CSE course, successfully completed under the honorable guidance of our course instructor and academic advisor, Prof. Dr. Mohammad Shahadat Hossain. Our team name is Epoch and our group number is 1.

This project was born from a simple but powerful goal: to protect communities from the dangers of flooding and unpredictable rainfall through real-time intelligent monitoring. Our system is a smart, IoT-integrated solution that uses live environmental sensing, multi-stage alert logic, and cloud connectivity to ensure timely warnings reach users before disaster strikes — even when no one is watching.

Bangladesh ranks among the world's most flood-vulnerable countries, with approximately 20–30% of national land flooded every monsoon season. Flash floods in particular often provide little to no warning time for affected communities. Our system addresses this critical gap through affordable, deployable IoT technology built on the ESP32 microcontroller and ThingSpeak cloud platform.

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2. Project Objectives

We set out to build a system that eliminates the uncertainty of flood monitoring for households, communities, and research institutions. By combining embedded hardware, real-time multi-sensor fusion, and IoT cloud connectivity through ThingSpeak, we empower users with a proactive, autonomous warning solution for modern environmental safety.

- To design and implement a low-cost IoT-based water level monitoring system using the ESP32 microcontroller
- To develop a 4-stage intelligent alert system that provides graduated warnings based on real-time water level measurements
- To integrate rain detection, temperature, and humidity sensing for comprehensive environmental monitoring
- To upload all sensor data to the ThingSpeak cloud platform for remote monitoring and historical analysis
- To demonstrate the system through a functional prototype using a controlled demo environment
- To evaluate system performance and establish a foundation for future large-scale river deployment

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3. Market Opportunity and Impact

Flooding and unexpected rainfall cause enormous damage to lives and property every year. In Bangladesh alone, floods affect millions of people annually. The problems our system addresses include:

- Loss of life and property due to delayed or absent flood warnings at the community level
- Expensive and complex traditional monitoring systems inaccessible to rural communities
- Lack of real-time data visibility for households and local authorities
- Dependence on centralized government systems that may not cover small rivers and local water bodies

Our system responds with the following capabilities:

- **Continuous automatic water level measurement** every 16 seconds using ultrasonic sensing
- **4-stage graduated alert system** with distinct buzzer and LED patterns for each danger level
- **Real-time cloud monitoring via ThingSpeak** for remote visibility from any device
- **Rain detection integration** to provide compound environmental awareness
- **Temperature and humidity monitoring** for complete environmental context
- **Low-cost hardware** deployable at under BDT 1,000 per sensor node

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4. System Components and Hardware

The following table lists all hardware components used in the system, along with their specific roles in the overall architecture.

Component	Specification	Role in System
ESP32 Development Board	30-pin, dual-core, Wi-Fi + BLE	Main controller — sensor reading, alert logic, Wi-Fi upload
HC-SR04 Ultrasonic Sensor	2–400 cm range, ±3mm accuracy	Primary water level measurement via time-of-flight
FC-37 Rain Sensor	Digital output only	Real-time rainfall detection — YES/NO binary output
DHT11 Sensor	Temp: 0–50°C, Humidity: 20–90%	Ambient temperature and humidity for environmental context
Active Buzzer (5V)	Active oscillator type	Audible staged alert — 1, 2, or 3 beep patterns
Red LED + 330Ω Resistor	5mm, red, 20mA	Visual alert indicator — blink or solid depending on stage
USB Powerbank	10,000 mAh	Portable power supply for prototype deployment
ThingSpeak Platform	Free tier, 4 fields	Cloud storage, time-series graphs, remote monitoring

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5. System Architecture and Working Principle

The system operates in a continuous loop, reading all sensors every 16 seconds and uploading data to ThingSpeak immediately after each reading cycle. The 16-second interval was deliberately chosen to stay safely above ThingSpeak's free-tier minimum rate limit of 15 seconds, eliminating upload rejection errors.

5.1 Water Level Measurement

The HC-SR04 ultrasonic sensor is mounted above the water container at a calibrated height of 16.0 cm above the empty container base. This calibration value was determined empirically by measuring the actual sensor-reported distance with the container completely empty, rather than relying on a tape-measure approximation. The sensor fires 5 ultrasonic pulses per reading cycle, sorts the resulting distance measurements using a bubble sort algorithm, and takes the median (third) value to eliminate noise caused by surface ripples or measurement inconsistency. Water level is then computed as:

Water Level (cm) = Sensor Height (16.0 cm) – Measured Distance to Water Surface

5.2 Four-Stage Alert System

The system divides the total container depth of 7 cm into four graduated alert stages, each with a distinct buzzer pattern and LED behaviour. The stage thresholds are calculated automatically as percentages of the total container depth:

Stage	Water Level Range	Threshold	Buzzer	LED	Meaning
1 — Normal	0.00 – 1.74 cm	< 25% depth	Silent	Off	No flood risk
2 — Watch	1.75 – 3.49 cm	25% – 49%	1 short beep	Off	Water rising, monitor closely
3 — Warning	3.50 – 5.24 cm	50% – 74%	2 beeps	Blinks 2×	Significant rise, prepare
4 — Danger	≥ 5.25 cm	≥ 75% depth	3 rapid beeps	Solid ON	Critical level, take action

When the water level drops back below Stage 2 threshold from any elevated stage, a single long all-clear beep sounds to explicitly confirm return to normal conditions. Alert beeps repeat every 15 seconds while elevated conditions persist.

5.3 Wi-Fi and Cloud Connectivity

The ESP32 connects to the local Wi-Fi network using the built-in Wi-Fi radio. The connection routine clears any stale state, attempts connection for up to 20 seconds, and retries up to 3 times at startup. A maintainWiFi() function runs before every upload cycle to automatically reconnect if the connection has dropped — ensuring uninterrupted data transmission during extended operation.

6. ThingSpeak Cloud Integration

All sensor data is uploaded to ThingSpeak channel ID 3413395 every 16 seconds. ThingSpeak automatically generates time-series graphs for each field, providing a complete historical view of water level changes, rainfall events, and environmental conditions over time.

Field	Data	Unit	Purpose
Field 1	Water Level	cm	Primary flood monitoring metric — main research data
Field 2	Rain Detected	0/1	0 = currently raining, 1 = dry conditions
Field 3	Temperature	°C	Ambient temperature for environmental context
Field 4	Humidity	%	Relative humidity for environmental context

The Alert Stage (1–4) is computed locally on the ESP32 from Field 1 data and displayed in the Serial Monitor. The ThingSpeak dashboard provides live charts and historical trend analysis accessible from any browser or mobile device.

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7. Hardware Wiring and Pin Connections

Component	Component Pin	ESP32 GPIO	Notes
HC-SR04	VCC	VIN (5V)	Powers sensor at 5V
HC-SR04	GND	GND	Common ground
HC-SR04	TRIG	GPIO 5	Trigger pulse output
HC-SR04	ECHO	GPIO 18	Via 1kΩ/2kΩ voltage divider — 5V→3.3V
FC-37	VCC	VIN (5V)	Powers rain sensor at 5V
FC-37	GND	GND	Common ground
FC-37	DO	GPIO 14	Digital rain detection only
DHT11	VCC	3.3V	Powered at 3.3V
DHT11	GND	GND	Common ground
DHT11	DATA	GPIO 4	10kΩ pull-up resistor to 3.3V
Buzzer	+	GPIO 13	Active buzzer — HIGH = ON
Buzzer	–	GND	Common ground
Red LED	Anode	GPIO 2	Via 330Ω current limiting resistor
Red LED	Cathode	GND	Common ground

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8. Key Achievements

- **Designed and implemented** an intelligent 4-stage water level alert system using ESP32 and HC-SR04 ultrasonic sensor
- **Integrated real-time cloud monitoring** using the ThingSpeak IoT platform with 4-field data upload every 16 seconds

- **Built a graduated alert mechanism** using buzzer and LED with distinct patterns for each of the four danger stages
- **Incorporated FC-37 rain sensor** for real-time rainfall detection alongside water level monitoring
- **Integrated DHT11 sensor** for ambient temperature and humidity data collection and cloud upload
- **Implemented robust Wi-Fi management** with automatic reconnection and 3-retry startup logic
- **Applied HC-SR04 median filtering** across 5 samples per reading cycle to eliminate noise
- **Successfully calibrated the system** to a 7 cm demo container with 4 alert stages at 25%, 50%, and 75% thresholds
- **Achieved reliable ThingSpeak uploads** with comprehensive error detection and diagnostic messaging for all common failure codes
- **Demonstrated the complete system** in a live prototype with real-time Serial Monitor output and ThingSpeak cloud graphs

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9. Results and Observations

The system was tested using a 7 cm deep container as a demo river environment. The following results were observed during prototype testing:

- Water level readings from HC-SR04 were stable and accurate within ±0.5 cm after median filtering
- All four alert stages triggered correctly at the expected water level thresholds
- Buzzer and LED patterns were clearly distinguishable for each stage during live demonstration
- ThingSpeak uploads succeeded consistently at 16-second intervals with no rate-limit rejections
- Wi-Fi auto-reconnection functioned correctly when connection was manually interrupted during testing
- DHT11 readings remained stable at approximately 24–27°C and 50–65% humidity throughout testing
- FC-37 rain sensor correctly reported DRY (1) under normal conditions and RAINING (0) when wetted
- Serial Monitor provided complete real-time diagnostic output confirming all sensor and upload operations

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10. Future Research Scope and Vision

We aim to make flood monitoring accessible, affordable, and intelligent for every community that needs it most. This prototype demonstrates that a sub-BDT 1,000 IoT node can provide meaningful early warning capability. Our future development roadmap includes:

- **Weather API integration** for predictive alerting based on upcoming forecast data
- **LSTM machine learning model** trained on historical water level data for flood prediction 6–24 hours in advance
- **Solar-powered autonomous nodes** for permanent off-grid riverbank deployment
- **LoRa mesh networking** for areas without Wi-Fi coverage, enabling kilometre-range communication
- **SMS and mobile push notifications** via IFTTT or GSM module for community-level alerts
- **Multi-node deployment** along entire river catchment areas with spatial data visualization
- **BWDB data integration** comparing system readings against official Bangladesh Water Development Board gauge stations
- **Publication as research paper** targeting IEEE Sensors Journal or Natural Hazards and Earth System Sciences

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11. Project Team

Name	Student ID	Role
Mohammad Ashraful Islam	240321022	Hardware setup, sensor wiring, firmware development
Md. Ali	230321003	Cloud integration, ThingSpeak setup, system testing

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